

# KONSTANTIN NIKOLAEV

## Software Engineer | Frontend and Full-Stack (React, TypeScript)

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### SUMMARY

Frontend-focused full-stack engineer who designs, builds and runs complex React and TypeScript web applications end to end. Sole engineer on three products with role-based access, encrypted secrets, OAuth 2.1 and automated accessibility tests. Built my foundations by hand from 2022 (data structures and algorithms, list virtualization, GraphQL) before adopting AI-assisted development, which I now use daily behind type checks, automated tests and my own review.

### TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript (ES6+), HTML, CSS, SQL, Rust

**Frontend:** React 18/19, Next.js (App Router), state management (TanStack Query, Zustand, Context), Tailwind CSS, Radix/shadcn

**Backend and Data:** Node.js, RESTful APIs, OpenAPI 3.1, GraphQL, SQLite/libSQL (Turso), MySQL, MongoDB, Drizzle, Prisma

**Security:** RBAC, signed session cookies, OAuth 2.1 with PKCE, TOTP two-factor, Argon2/scrypt, AES-256-GCM, CSP, rate limiting

**Testing and Delivery:** Vitest, Playwright, axe-core (WCAG 2.2 AA), Git, Vercel

**AI-Assisted Development:** Claude Code, Anthropic API (streaming, tool use), MCP servers, RAG

### PROFESSIONAL EXPERIENCE

#### Full-Stack Software Engineer, Agency Collective

Nov 2025 – Present | Remote

*Sole engineer at a performance marketing agency. I own architecture, delivery, security and documentation, working directly with sales, finance and operations.*

**Agency Collective Dashboard** | Next.js, React, TypeScript, TanStack Query, SQLite (Turso) | admin, client and sales portals

- **Deal creation and review:** built the pipeline closers, setters and admins use daily. One form creates the deal with its draft invoice and e-sign contract and feeds commission and payout records. Approval claims the draft and inserts the deal in one atomic libSQL batch (compare-and-set on the reviewed version, 409 on stale edits), over a schema with 70 indexes (partial unique, covering).
- **E-signature:** replaced DocuSign with open-source DocuSeal to remove per-envelope cost: embedded React template builder and in-portal signing, a Zod-validated API client with retries, webhook status sync and signed PDFs filed to the payout ledger.
- **REST API and MCP server:** designed 180+ operations with a hand-written OpenAPI 3.1 spec, scoped tokens hashed at rest and per-token rate limits, plus an MCP server that generates one tool per operation behind OAuth 2.1 with PKCE. The company's AI agents use it daily to query and analyze clients, deals and billing, and their writes land as drafts for human approval.
- **Client management:** multi-role React app (56 pages, 270+ components) with 13-permission RBAC enforced in edge middleware and again in every route, HMAC-signed sessions and row-level workspace scoping for partner teams. A unit-tested re-bill engine reconciles invoices with payments.
- **Team task manager (Asana-style):** per-member Kanban boards with optimistic TanStack Query updates and server-computed ordering. An AI agent reads Slack and files tasks and action items through the MCP server.

**RUO Commerce** | Next.js 16, React 19, Zustand, Drizzle, Stripe | Shopify-style storefront and admin

- **Commerce back end:** 76-table Drizzle schema covering catalog with lot-level lab certificates, a pricing engine (volume tiers, discounts, gift cards, loyalty), Stripe behind a payment-provider abstraction, tax, shipping, 3PL fulfillment and refunds with restock. Pages prerender with tag-based cache invalidation at one database request per render, and server-side Meta and TikTok conversion events deduplicate against browser pixels by event ID.
- **MCP server with 79 tools:** the store is its own OAuth 2.1 authorization server, and agents act under the connecting staff member's role and scopes to manage products, pages, design and orders. High-impact calls return a preview and need a single-use HMAC confirmation token bound to the arguments and current state. Compare-and-set drafts keep agents and the visual editor from overwriting each other.
- **Visual editor and quality:** drag-and-drop sections with a live preview iframe driven by a typed postMessage protocol, undo and draft/publish. Argon2 and TOTP two-factor sign-in, AES-256-GCM for stored secrets, 200+ unit test files, and Playwright suites with axe WCAG 2.2 AA checks on desktop and mobile.

**PeptideAds Assistant** | Next.js 15, React 19, Anthropic API | invite-only AI chat product

- Built a streaming chat interface and agent tool loop with retrieval over a curated knowledge base (hybrid vector and full-text search on SQLite), per-user spend caps metered to the cent and prompt-injection fencing of untrusted content.

### ENGINEERING FOUNDATIONS (independent work, 2022 – 2024, before AI-assisted development)

- **List virtualization in React:** windowing hooks for fixed and dynamic lists and 2D grids, plus debounce and throttle, no libraries.
- **Real-time messenger:** Next.js, Apollo GraphQL subscriptions over WebSockets, and a Node.js server on Prisma and MongoDB.
- **Multi-store e-commerce dashboard and storefront:** Next.js, TypeScript, Prisma, MySQL and Stripe checkout.
- **Data structures and algorithms:** heaps, hash tables, trees, weighted graphs (Dijkstra) and a shunting-yard expression parser.

### EDUCATION

California State University, Northridge | B.A. Economics and B.S. Management, Cum Laude | GPA 3.65

2017 – 2021